

SRNN model equations

Preset: celltype_pairs_Sc0p2_noise0p025_dualStd_7cond · **Model class:** SRNNCellTypePairs

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System equations

$$dx_i = \frac{-x_i + u_i + \sum_j w_{ij} r_j \prod_{m=1}^2 b_{jm}}{\tau_d} dt + \frac{\sigma_u}{\tau_d} dW_i$$

Additive Wiener noise enters **only** x , which keeps the diffusion constant: Ito and Stratonovich coincide, the Milstein term vanishes, the variational equation is untouched, and the noise cancels in Benettin’s trajectory difference. σ_u is input-referred, so it is comparable to the stimulus amplitude.

$$r_i = \phi \left(x_i - a_{0_i} - \frac{c}{K} \sum_{k=1}^K a_{ik} \right)$$

ϕ is centred on zero; a_{0_i} is the **setpoint** (`S_c`, 0.2 here), per neuron when `mu_S_c` / `sigma_S_c` are set.

The rate r_i is the **pre-depression** output of the nonlinearity. Depression and facilitation enter presynaptically, as factors multiplying r_j inside the recurrent sum, so both mechanisms are driven by the raw rate r_i and not by the depressed output.

$$\begin{aligned} \frac{da_{ik}}{dt} &= \frac{-a_{ik} + r_i}{\tau_{a_k}} \\ \frac{db_{im}}{dt} &= \frac{1 - b_{im}}{\tau_{rec_m}} - \frac{b_{im} r_i}{\tau_{rel_m}}, \quad m = 1, \dots, 2 \\ \theta_j &= r_j \prod_{m=1}^2 b_{jm} \end{aligned}$$

θ_j is the **synaptic output**: the presynaptic rate after depression, and the quantity the recurrent sum above transmits.

The depression timescales enter as a **product**, which is the reason for using more than one: 2 of them SQUARE the depression at steady state, so the extra timescale buys DEPTH and not merely a slower approach. There is no split of τ that would make them match a single timescale.

With $\tau_{rec}/\tau_{rel} = 8$ on every timescale, each factor relaxes to $b_k(r) = 1/(1 + 8r)$, so at $r = 1$ a single timescale gives a gain of 0.1111 (a 9-fold reduction against an undepressed synapse) while 2 give 0.01235 – a **81-fold** reduction, exactly the square.

Adaptation, by contrast, enters as a **sum**, and is normalised by its timescale count: the rate subtracts $(c/K) \sum_{k=1}^K a_k$. Since every a_k relaxes to the rate, $\sum_k a_k \rightarrow Kr$ regardless of the τ_k , so the steady state is cr whatever K is – adding SFA timescales changes the route, not the destination. c is therefore the **total** adaptation budget. No such normalisation applies to depression, which multiplies rather than sums.

The sparse weights are drawn element-wise and scaled by the synaptic gain g :

$$w_{ij} = g S_{ij} (\tilde{\sigma}_{ij} \xi_{ij} + \tilde{\mu}_{ij}), \quad \xi_{ij} \sim \mathcal{N}(0, 1), \quad S_{ij} \sim \text{Bernoulli}(\alpha)$$

with $(\tilde{\mu}, \tilde{\sigma})$ indexed by (**postsynaptic**, **presynaptic**) cell type, and given as multiples of $F = 1/\sqrt{n\alpha(2-\alpha)}$.

Parameters as run

Symbol	Name	Value	Units
n	Network size	500	neurons
α	Connection probability	0.2	—
indegree	Expected in-degree	100	synapses
f	Fraction per cell type	[0.5 0.5]	—
τ_d	Dendritic time constant	0.1	s
g	Synaptic gain	1	—
$\tilde{\mu}$	Weight means (post, pre)	[5.5 -5.5; 5.5 -5.5]	multiples of F
$\tilde{\sigma}$	Weight std devs (post, pre)	[1.5 1.5; 1.5 1.5]	multiples of F
ϕ	Nonlinearity	piecewise	—
S_c	Nonlinearity setpoint	0.2	—
S_a	Piecewise slope parameter	0.8	—
K	SFA timescales per type	[3 0]	—
τ_a (E)	SFA time constants	[0.25 1.581 10]	s
c	SFA coupling per type	[0.5 0]	—
τ_{rec} (E to E)	STD recovery	[2 4]	s
τ_{rel} (E to E)	STD release	[0.25 0.5]	s
τ_{rec} (E to I)	STD recovery	[2 4]	s
τ_{rel} (E to I)	STD release	[0.25 0.5]	s
τ_{rec} (I to E)	STD recovery	[2 4]	s
τ_{rel} (I to E)	STD release	[0.25 0.5]	s
τ_{rec} (I to I)	STD recovery	[2 4]	s
τ_{rel} (I to I)	STD release	[0.25 0.5]	s
σ_u	Input-referred noise	0.025	input units
σ_x	Stationary std of x	0.0559017	—
f_s	Sample rate	400	Hz